Name: ______________________

VECTORS, (VECTOR GEOMETRY PROOFS)

Date: ________________

LO: I can use vectors to prove geometric results involving midpoints, parallel lines, and collinear points.

Vector geometry proofs

In **vector geometry**, we express journeys around a shape in terms of vectors **a** and **b**. If two vectors are **parallel** (scalar multiples), the lines are parallel.

Key strategies: To prove collinearity, show one vector is a scalar multiple of another AND they share a common point. The midpoint of AB has position vector $\frac{1}{2}$(a + b).

Quick examples

●	AB = b - a (position vectors)	from A to B
●	Midpoint M of AB: OM = $\frac{1}{2}$ (a + b)	average of position vectors
●	If PQ = kRS, then PQ is parallel to RS	scalar multiple means parallel
●	AB = a + b	should be b - a
●	Parallel vectors prove	also need a shared

Key idea

Express unknown vectors as journeys using known vectors. Parallel means scalar multiple; collinear needs shared point too.

AB = b - a

Travel from A to O (which is -a), then O to B (which is +b)

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – expressing a vector

OA = a, OB = b. Express AB in terms of a and b.

AB = AO + OB

AO = -a (reverse of OA)

AB = -a + b = b - a

AB = b - a

Go backwards along a, then forwards along b.

Example 2 – finding a midpoint

M is the midpoint of AB. OA = a, OB = b. Find OM.

OM = OA + AM

AM = $\frac{1}{2}$AB = $\frac{1}{2}$(b - a)

OM = a + $\frac{1}{2}$(b - a) = $\frac{1}{2}$(a + b)

OM = $\frac{1}{2}$(a + b)

The midpoint is the average of the position vectors.

Example 3 – proving lines are parallel

OA = a, OB = b. M is midpoint of OA, N is midpoint of OB. Show MN is parallel to AB.

OM = $\frac{1}{2}$a, ON = $\frac{1}{2}$b

MN = -$\frac{1}{2}$a + $\frac{1}{2}$b = $\frac{1}{2}$(b - a)

AB = b - a, so MN = $\frac{1}{2}$AB. Parallel.

MN = $\frac{1}{2}$AB, so parallel

MN is a scalar multiple of AB, proving they are parallel.

Example 4 – proving collinearity

OA = a, OB = 3a. Show O, A, B are collinear.

OA = a and OB = 3a

AB = OB - OA = 3a - a = 2a

OA = a and AB = 2a: AB = 2(OA),

same direction, shared point A

O, A, B are collinear

AB is a scalar multiple of OA and they share point A.

YOUR TURN – PRACTICE (deliberate practice)

1. Expressing vectors

OA = a, OB = b. Express in terms of a and b.

a) AB

b) BA

c) 2OA + OB

d) OA + $\frac{1}{2}$AB

2. Midpoints

Find the position vector of the midpoint M.

a) OA = (2, 6), OB = (4, 8). M midpoint of AB.

b) OA = (1, 3), OB = (5, 7). M midpoint of AB.

c) OA = a, OB = b. M midpoint of AB.

d) OA = (0, 4), OB = (6, 0). M midpoint of AB.

3. Proving parallel lines

Show that the vectors are parallel.

a) PQ = 2a + b, RS = 6a + 3b

b) AB = a - 2b, CD = 3a - 6b

c) EF = 4a + 2b, GH = 2a + b

d) MN = -a + 3b, PQ = 2a - 6b

4. Ratio problems

P divides AB in the given ratio. Find OP.

a) OA = a, OB = b. P divides AB in ratio 1:1.

b) OA = a, OB = b. P divides AB in ratio 1:2.

c) OA = a, OB = b. P divides AB in ratio 2:1.

d) OA = (2, 1), OB = (8, 7). P divides AB in ratio 1:2.

5. Mixed vector proofs

Solve each problem.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) AB = b - a when OA = a and OB = b ☐

Correct answer: __________

Reason: __________________

b) AB = a + b ☐

Correct answer: __________

Reason: __________________

c) If PQ = kRS, then PQ and RS are parallel ☐

Correct answer: __________

Reason: __________________

d) Parallel vectors always prove collinearity ☐

Correct answer: __________

Reason: __________________

e) Midpoint of AB = $\frac{1}{2}$(a + b) when OA = a, OB = b ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) In triangle OAB, OA = **a** and OB = **b**. M is the midpoint of OA, N is the midpoint of AB. Prove that MN is parallel to OB and find the ratio MN : OB.

Expression: ______________________________

Simplified: ______________________________

b) OABC is a quadrilateral with OA = **a**, OC = **c**, and AB = **c**. Show that OABC is a parallelogram. M is the midpoint of OB and N is the midpoint of AC. Prove that M and N are the same point.

Expression: ______________________________

Simplified: ______________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!